PUF-CLGS: Application of Multi-Factor PUF Authentication Circuit Based on Lattice Group Signature in Drone-Assisted Communication

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***Abstract*—In drone-assisted communication scenarios, especially in post-disaster rescue, there is a risk of privacy leakage when heterogeneous drone groups collaborate. Therefore, ensuring the anonymity of drones while guaranteeing the reliability of the message source is a significant challenge. Although existing schemes achieve anonymity, they are unable to dynamically adjust the authentication difficulty according to changes in the physical environment. In this paper, we propose an application of multi-factor puf authentication circuit based on lattice group signature for in drone-assisted communication. The scheme innovatively designs a PUF-based multi-factor authentication circuit and proposes a dynamic authentication strategy, achieving authentication and physical security under different security requirements. The scheme proposes a lattice-based heterogeneous drone group signature scheme through Merkle hash tree and lattice cryptography, which achieves efficient anonymous authentication and group member update and ensures post-quantum security. The security of the scheme is proved by formal security analysis based on AVISPA and random oracle model (ROM). Moreover, experimental results show that PUF-CLGS outperforms the benchmark schemes in terms of both computational and communication costs.**

***Index Terms*—Heterogeneous drone, drone swarm, dynamic authentication, group signature, lattice-based cryptography, physical unclonable function (PUF) circuit.**

I. Introduction

In RECENT years, natural disasters have had a significant impact on life and economy around the world. The Centre for Research on the Epidemiology of Disasters (CRED) reported that in 2023, there were 399 natural disasters, resulting in 86,473 deaths and economic losses up to $202.7 billion [1]. Ground-based infrastructure (e.g., roadside communication units) is often damaged during disasters, hindering the transmission of rescue information and reducing rescue efficiency. The Internet of Drones (IoD) has emerged as a key solution due to its rapid deployment capability [2].

Due to their low cost, ease of deployment, and high flexibility [3], drones can quickly collect and transmit information about disaster areas, enabling efficient rescue. However, in post-disaster rescue, IoD systems face the following challenges: Firstly, drones are vulnerable to physical cloning attacks due to their high flexibility, which pose a risk to identity authentication. Secondly, drones in an open environment are vulnerable to tampering or forgery during data transmission, thus threatening the reliability of rescue. Then, according to the different types of rescue tasks, the rescue drones are categorized into groups such as life search and rescue, supply delivery, and medical support. Each group employs different communication protocols, such as the life search and rescue group using the MAVLink protocol, and the supply delivery group using the LoRaWAN protocol. Since the collaboration between these heterogeneous drones poses a risk of privacy leakage, it is crucial to ensure the anonymity of the drones. However, to prevent malicious nodes from spreading false information, we need to design an authentication scheme that can ensure anonymity and achieve traceability.

Existing authentication schemes are mainly divided into traditional authentication and group authentication. Traditional authentication uses digital certificates to verify the identity of individual devices, but it incurs high computational costs. Group authentication verifies whether an entity belongs to a group, but it lacks traceability. However, group signature is a specialized form of group authentication that allows group members to generate verifiable anonymous signatures, ensuring data integrity while providing conditional privacy protection. Therefore, group signature is more suitable for achieving anonymous authentication and traceability.

With the development of quantum computers, cryptographic methods based on classical number theory are being challenged by quantum computing. Lattice-based cryptography [4], due to its quantum-resistant properties and computational efficiency, has become a research hotspot in post-quantum cryptography. In 2024, National Institute of Standards and Technology (NIST) [5] has included lattice-based cryptography in the post-quantum cryptography standardization framework, providing new security guarantees for cybersecurity. The security of lattice-based cryptography relies on its theoretically hard problems, such as the Learning With Errors

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TABLE I
SUMMARY OF EXISTING WORK

Scheme	Methods	Limitations/Challenges
[7],[8]	PKC	The use of public-key cryptography leads to high computational overhead
[9]	Symmetric encryption	Symmetric encryption offers low computation overhead, but poor scalability
[10],[11]	Blockchain	The use of blockchain results in high communication overhead and makes the system vulnerable to physical cloning attacks
[13]-[15]	PUF	Employs single-factor authentication with poor scalability and lacks adaptability to varying security requirements across scenarios
[17]-[22]	Classical number theory	High computational overhead and inability to resist quantum attacks
[24]-[26]	Lattice cryptography	Resists quantum attacks, but lacks flexibility and is vulnerable to physical cloning attacks

(LWE) problem, for which no efficient quantum algorithms are currently known. Since it only requires matrix and vector operations, it is well-suited for resource-constrained drones.

A. Motivation

In post-disaster, the flexibility and rapid deployment capabilities of IoD have proven to be of great value. However, to fully realize its potential, several issues still need to be addressed. Firstly, existing schemes often overlook the possibility of drones being subjected to physical cloning attacks, which allows attackers to forge drone identities by cloning devices and poses a threat to system security. Therefore, it is essential to achieve drone identity authentication along with hardware tamper-resistance. Secondly, existing single-factor or multi-factor authentication strategies are difficult to adapt to the changing security requirements of different scenarios, resulting in low verification efficiency. Therefore, it is necessary to design a dynamic authentication strategy that adapts to different security requirements. Moreover, during the collaboration of heterogeneous drones, there is a risk of privacy leakage, and anonymity may be maliciously exploited by attackers to spread false information. Therefore, it is necessary to strike a balance between anonymity and traceability. However, group signature schemes based on classical number theory are vulnerable to quantum computing attacks. Thus, it is necessary to design a lattice-based group signature scheme to resist such threats.

B. Contribution

Based on the above discussion, this paper proposes an application of multi-factor PUF authentication circuit based on lattice group signature in drone-assisted communication, achieving both post-quantum security and physical security. We also conducted a comprehensive security analysis and evaluated the proposed scheme through experiments. The main contributions are as follows:

1) We are the first to jointly exploit the nonlinear characteristics of physical-layer responses and logical-layer strategies to innovatively design a PUF-based multi-factor authentication circuit. Building on this, we propose a PUF-based dynamic authentication strategy to meet diverse security requirements and achieve both dynamic authentication and physical-layer security.
2) We propose a lattice-based group signature scheme for heterogeneous drones by combining Merkle hash tree with lattice cryptography, which enables efficient anonymous authentication and group membership updates. As the scheme is constructed based on the ring learning with errors (RLWE) problem, it ensures post-quantum security.
3) We verify the security of the scheme through a Random Oracle Model (ROM) and informal analysis, and performed formal verification using the AVISPA tool to prove that the scheme meets the predefined security requirements. Experimental results show that PUF-CLGS not only meets all security requirements but also significantly reduces computational overhead and communication cost compared to the baseline scheme.

C. Article Organization

The rest of this article is organized as follows. Section II introduces related work. Section III presents the background on the associated techniques. Section IV describes the system model, threat model, security goals, signature framework, and security model. Section V outlines the proposed scheme. Sections VI and VII are devoted to security verification and analysis, and experimental study, respectively. Section VIII summarizes our work. Section IX discusses our future work.

II. RELATED WORK

This section reviews existing identity authentication and group signature schemes, analyzing and summarizing their methods and limitations of existing schemes, as shown in Table I for details.

A. Identity Authentication

With the advent of the Internet of Everything era, identity authentication has become the cornerstone of information security [6]. In post-disaster rescue with limited resources, it is critical to realize secure and reliable identity authentication.

In recent years, most research on authentication has been based on traditional cryptographic methods such as public key cryptography (PKC) and symmetric encryption. Authentication schemes based on PKC [7], [8] incur high computational overhead due to the use of bilinear pairing or elliptic curve cryptography, whereas those based on symmetric encryption [9] are more efficient but lack scalability.

The researchers also proposed lightweight authentication schemes based on blockchain. Shen et al. [10] designed a batch authentication scheme for vehicular networks using a certificateless mechanism without bilinear pairing, while Luo et al. [11] proposed an efficient authentication scheme by integrating blockchain with fog computing, enhancing efficiency through hierarchical blockchain and parallel consensus. Although these

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schemes reduce computational costs, they fail to consider the vulnerability of devices to physical cloning attacks.

Hardware-based identity authentication offers a promising approach to addressing the aforementioned issues. PUF [12] leverage the inherent randomness of hardware microstructures for authentication, providing a natural security advantage due to the non-replicable nature of their physical characteristics.

Zhang et al. [13] proposed an AKA scheme combining chameleon hash function and PUF, which features low computational costs but poses a risk of privacy leakage. Mexis et al. [14] designed an authentication scheme based on DRAM PUF and machine learning. Although the computational complexity is low, it only uses single-factor authentication and has poor flexibility. Pu et al. [15] proposed an group authentication protocol for IoD, which achieves drones group authentication through lightweight operations, but does not consider defense against malicious nodes.

Although PUF-based authentication schemes [13], [14], [15] can resist physical cloning attacks, they are mostly deployed in conventional environments and use single-factor authentication, without considering that conventional authentication methods may lead to a high device rejection rate in post-disaster rescue. If the authentication conditions are relaxed to improve the success rate, the device security may be compromised. Thus, it is necessary to design a dynamic authentication strategy to achieve the authentication of drones under different safety requirements.

B. Group Signature

Researchers have extensively studied group signature schemes [16], but most are based on traditional cryptographic methods. Zhang et al. [17] proposed a message-linkable group signature scheme that achieves anonymity but incurs high computational costs. Scheme [18] utilized Merkle hash tree to enable one-time authentication for multiple accesses, reducing authentication overhead. Bao et al. [19] designed a group signature scheme with selective linkability and traceability, enabling fine-grained and efficient access control. Cai et al. [20] proposed a group signature scheme that supports batch verification, improving authentication efficiency. Although these schemes address the anonymity issue, they cannot resist quantum computing attacks due to their reliance on classical number theory [21], [22].

Lattice security relies on difficult lattice problems (such as LWE) [23], which remain quantum-resistant even in quantum computing environments. Thus, researchers have proposed lattice-based schemes [24], [25], [26]. For example, Prajapat et al. [24] proposed a lattice-based aggregate signature scheme, which achieves efficient authentication through signature aggregation but does not consider physical clone attacks. Tang et al. [25] proposed a lattice-based event-driven linkable group signature scheme, which is suitable for scenarios with a limited number of signatures but lacks flexibility.

In summary, compared with existing schemes, this scheme achieves the anonymity and flexible management of drones, while also providing resistance to quantum computing and physical clone attacks, enhancing security. In addition, this scheme has lower computational overhead, enhancing the practicality of the system.

TABLE II
NOTATIONS IN THE PROPOSED SCHEME

Notation	Description
N/a	Group size / Depth of the tree
$\mathrm{gpk}/\mathrm{gsk}$	Group public key / Group private key
$\mathrm{epk}/\mathrm{esk}$	CD's public / private key
$\mathrm{tpk}/\mathrm{tsk}$	TM's public / private key
$\mathrm{info}_\tau/\pi_{tr}$	Group information at period τ / Trace proof
RD_i	The identity of the i-th RD
CD_{id}	The identity of CD
C_i	The i-th challenge
$\mathrm{C}_{(i,n)}/\mathrm{R}_{(i,n)}$	The n-th challenge / response of the i-th device
$\|\|$	Concatenation operation
SSC/SRC	CD-CS shared key / RD-CD shared key
$pp/r_k/r_{k_1}$	Public parameter / Random Numbers
$\mathrm{dpk}_i/\mathrm{dsk}_i$	RD_i's public / private key
reg/S	Registration list / Revocation List
$m/C/\sigma$	Message / Ciphertext / Group Signature
$T_i(i=0,1)/K$	Timestamp / Symmetric Key
X_{CS}/x	CS's public / private key
$H(\cdot)/H_1(\cdot)$	Hash function
$X_{CD}/Y_{CD}/Z_{CD}$	Authentication credentials
$SCD_{id}/CR/SC_2$	Authentication code between CS and CD
$RR/SC/RC$	Authentication code between CD and RD

III. PRELIMINARIES

In this section, we mainly introduce the techniques used. The main notations used in this paper are shown in Table. II.

A. Lattice

A lattice L is a discrete subgroup of the real vector space $\mathbb{R}^m$, generated by a set of linearly independent vectors (called a basis) [23]. Formally, given a set of lattice basis vectors $B = \{\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_n\}$, the lattice L is defined as:

$$L(B) = \left\{ \sum_{i=1}^{n} x_i \mathbf{b}_i \mid x_i \in \mathbb{Z} \right\}. \tag{1}$$

Here, x_i are integer coefficients, $\mathbb{Z}$ is the set of integers, and m and n represent the dimension of the vector space and the number of basis vectors, respectively.

B. Physical Uncloneable Function

PUF is a security technology that generates identifers based on microscopic differences in the chip manufacturing process [14], [27]. It generates a response R = PUF(C) based on the received challenge C. It has the following characteristics:

1) Unpredictability: The response R generated by the same challenge C is random and unpredictable, preventing attackers from guessing the response.
2) Uniqueness: Each PUF is unique, ensuring that the same challenge C produces different R for different PUFs.
3) Stability: PUF can generate the same responses to the same challenge under different environmental conditions.
4) Tamper Resistance: PUF characteristics are determined by subtle manufacturing variations, making them difficult to replicate. Any invasive attack would damage its structure and leave evidence of tampering [28].

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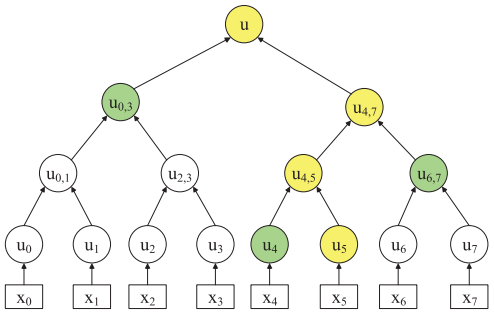


Fig. 1. Merkle Hash Tree.

C. PUF Type

PUF are classified based on manufacturing processes and security. According to the manufacturing process, they are divided into silicon PUF (e.g., Arbiter PUF [29]) and non-silicon PUF (e.g., Optical PUF [30]). Based on security, PUFs are classified as weak, strong, or controlled.

1) Weak PUF: Due to the limited challenge-response pair (CRP) space, it is typically used for key generation and digital fingerprints, such as SRAM PUF [31].
2) Strong PUF: With a larger CRP space, it is suitable for authentication and key establishment, such as Ring Oscillator PUF [32] and Arbiter PUF.
3) Controlled PUF: A strong PUF with additional control logic circuits to enable more functionalities.

D. Merkle Hash Tree

The Merkle hash tree computes leaf node values using a collision-resistant hash function and recursively calculates hashes up to the root node. Leaf nodes are verified using the root node value u and witness information w. As shown in Fig. 1, given N ($N = 8$) nodes, compute their hash values as $u_i = H(x_i)$, recursively calculate the parent node values up to the root. The group member sends u_5 and $w = (u_4, u_{0,3}, u_{6,7})$ to the CD. The CD computes $u_{4,5} = H(u_4 \| u_5)$, $u_{4,7} = H(u_{4,5} \| u_{6,7})$, and $u^* = H(u_{0,3} \| u_{4,7})$. It compares u^* with u to complete the verification. When revoking a member, the corresponding leaf node u_5 is set to 0. Only the values of the yellow nodes are updated, and the witness information w of other members is updated.

E. Complexity Assumptions

Let $R = \mathbb{Z}[X]/(X^n + 1)$ and $R_q = \mathbb{Z}_q[X]/(X^n + 1)$ denote the polynomial rings over the integers $\mathbb{Z}$ and the integers modulo q of $\mathbb{Z}_q$, respectively, where n is a power of 2 and q is a prime number. The norms L_2 and L_∞ are defined as $\|a\| = \sqrt{\sum_{i=0}^{n-1} a_i^2}$ and $\|a\|_\infty = \max |a_i|$, where $a = a_0 + a_1 X + \cdots + a_{n-1} X^{n-1} \in R$. Let χ be the discrete Gaussian distribution on R_q [33].

Definition 1 (RLWE Problem): Let $q \geq 2$ be a prime, and let R, R_q, and χ be defined as above. The RLWE distribution $\mathcal{A}_{s,\chi}^{\mathrm{RLWE}}$ is defined as $\mathcal{A}_{s,\chi}^{\mathrm{RLWE}} = (a, b = a \cdot r + e) \in R_q \times R_q$, where $a \leftarrow R_q$ and $s, e \leftarrow \chi$. Even with polynomial-time computational power, an adversary cannot effectively distinguish $\mathcal{A}_{s,\chi}^{\mathrm{RLWE}}$ from the uniform distribution over $R_q \times R_q$.

Definition 2 (ECDLP Problem): Let P, $Q \in E$, where P is the base point, $k < p$, and $Q = k \cdot P$. The ECDLP problem states that, given the points P and Q, it is computationally hard to determine k.

F. Digital Signature Scheme

A digital signature scheme consists of three algorithms: 1) $\mathrm{DSKg}(1^\theta) \rightarrow (sk, vk)$: Outputs key pair (sk, vk). 2) $\mathrm{DSSig}(sk, m) \rightarrow \sigma_{\mathrm{DS}}$: Generates the signature σ_{DS} for message m using the private key sk. 3) $\mathrm{DSVf}(vk, m, \sigma_{\mathrm{DS}}) \rightarrow \{0, 1\}$: Verifies whether $(m, \sigma_{\mathrm{DS}})$ is valid under vk. if valid, returns 1. otherwise 0. We adopt existential unforgeability under chosen message attack as the security of the scheme.

IV. System Model

This section introduces the system model, discusses the threat model and security goals, and elaborates on the syntax and security model of the scheme.

A. System Model

There are four main entities in system, as shown in Fig. 2.

1) Communication Drone (CD): CD serves as a temporary communication node in disaster areas, equipped with high data storage and processing capabilities. It generates group parameters, public-private key pairs for itself and TM, and stores the PUF database for RD authentication.
2) Rescue Device (RD): RD is equipped with multiple PUFs to meet authentication needs and has storage and encryption functions. It generates public and private keys for itself. Since RD is not trusted, identity verification is required before communication can be established.
3) Trace Manager (TM): TM is responsible for preventing RD from abusing anonymity. If necessary, TM can use its private key tsk to decrypt the ciphertext C, retrieving the malicious RD's identity and the tracking proof π_{tr}.
4) Cloud Server (CS): CS has powerful computing and storage capabilities, generates system parameters and its own public and private keys, stores CD identity and RD rescue information, verifies the legitimacy of CDs, and assigns rescue tasks to RDs.

B. Threat Model and Security Goals

The multi-party information exchange of IoD devices faces privacy and reliability threats. This paper defines three typical attack scenarios and makes reasonable assumptions about the attacker's capabilities.

1) Impersonation Attacks: The attacker forges the identity of a legitimate drone and sends false information to the system, affecting rescue coordination.
2) Eavesdropping Attacks: The attacker intercepts private data by exploiting insecure communication channels, potentially leading to device control or mission leakage.
3) Collusion Attacks: The attacker controls some members to collude in sending false data or misguiding routes, potentially exposing the signer's identity, compromising anonymity, and affecting rescue operations.

To address the above security threats, our scheme must meet the following goals:

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1) Correctness: PUF-CLGS ensures that signatures generated by honest signers are always accepted.
2) Anonymity: PUF-CLGS must ensure that only trusted CD know the real identity of the signer, and adversaries cannot leak the signer's true identity through eavesdropping.
3) Message Integrity: PUF-CLGS must authenticate the source of messages and ensure that the messages have not been subject to unauthorized manipulation or modification.
4) Non-frameability: PUF-CLGS must ensure that attacker cannot forge valid signatures or proofs acceptable to judges, and cannot attribute signatures to innocent users.
5) Traceability: PUF-CLGS must ensure the ability to reveal the identity of malicious drones when necessary.
6) Resistance Quantum Computing Attacks: PUF-CLGS must be guaranteed to resist quantum computing attacks.
7) Resistance MITM Attacks: PUF-CLGS must ensure that when the verifier verifies the RD signature, it confirms that the signature is generated by a legitimate device and the message has not been tampered with.
8) Resistance Collusion Attacks: PUF-CLGS must ensure that colluders using the same information in different authentications are identified by the Trace algorithm.

C. Signature Frame

Our proposed scheme consists of the following algorithms:

1) ENGen$(1^\theta, N) \rightarrow (pp, gpk, esk, tsk, info)$: Input security parameter θ and group size N, output public parameter pp, group public key gpk, CD's private key esk, TM's private key tsk and group information $info$.
2) DKGen$(1^\theta) \rightarrow (dpk_i, dsk_i)$: Input security parameter θ, output the public-private key pair (dpk_i, dsk_i) of the RD_i.
3) Join$(info_\tau, gpk, dsk_i) \leftrightarrow$ Issue$(info_\tau, esk, dpk_i) \rightarrow (gsk_i, reg)$: An interactive algorithm run by RD_i and CD. RD_i obtains the group private key gsk_i and CD adds each RD_i's identity information to the registration list reg.
4) GUpdate$(gpk, esk, info_\tau, S, reg) \rightarrow info_{\tau_{new}}$: CD revokes group member RD_i by calculating and updating group information $info_{\tau_{new}}$.
5) Sign$(gpk, gsk_i, info_\tau, C) \rightarrow \sigma$: First lattice-encrypt the message m, then use the group private key gsk_i to sign the ciphertext C. If RD_i is inactive at epoch τ, output $\perp$.
6) Verify$(gpk, info_\tau, C, \sigma) \rightarrow \{0, 1\}$: Verify the validity of the signature σ on the ciphertext C at epoch τ. If valid, output 1; otherwise, output 0.
7) Trace$(gpk, tsk, info_\tau, C, \sigma, reg) \rightarrow (RD_i, \pi_{tr})$: Given the input, if TM successfully traces, output the signer's identity RD_i and the trace proof π_{tr}, otherwise output $\perp$.
8) Judge$(gpk, RD_i, info_\tau, C, \sigma, \pi_{tr}) \rightarrow \{0, 1\}$: Verify the validity of the trace proof π_{tr} output by the Trace algorithm. If valid, output 1; otherwise, output 0.

D. Security Model

We define a series of oracles to meet the different query requirements of the adversary $\mathcal{A}$ in various experiments.

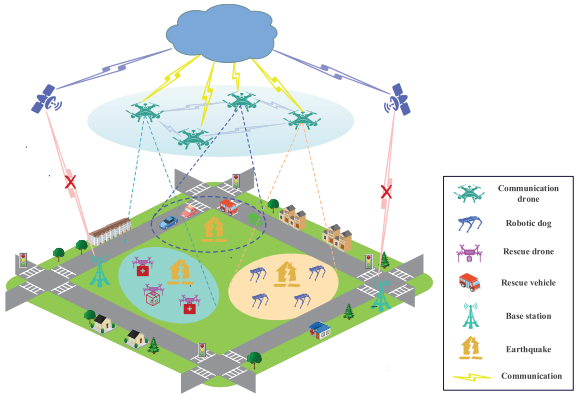


Fig. 2. System Model.

AddD(RD_i): The new honest device RD_i executes this oracle and joins the group. Then add it to the list HDL, and this oracle returns dpk_i.

CrptD(RD_i, dpk): $\mathcal{A}$ uses this oracle to corrupt a RD_i and set its public key to dpk_i. This oracle adds RD_i to the list CDL.

SndToEN(RD_i, M_{in}): This oracle participates in the $\langle$Join, Issue$\rangle$ protocol between the honest CD and the adversary controlling the corrupted device $\mathrm{RD}_i \in$ CDL.

SndToD(RD_i, M_{in}): This oracle participates in the $\langle$Join, Issue$\rangle$ protocol between the honest device $\mathrm{RD}_i \in$ HDL and the adversary controlling the corrupted CD.

Reveal(RD_i): The adversary $\mathcal{A}$ uses the oracle to obtain the private key dsk_i and group private key gsk_i of the device RD_i. Then, the oracle adds it to the BUL list.

RReg(RD_i): The adversary $\mathcal{A}$ uses the oracle to retrieve the reg_i related to the device RD_i.

WReg(RD_i, ρ): The adversary $\mathcal{A}$ can use this oracle to write a random string ρ into the reg_i associated with the device RD_i.

EUpdate(S): The adversary $\mathcal{A}$ can use this oracle to revoke the device RD_i. Then, the oracle updates the list reg_i and adds its public key to the revocation list S.

Sign(RD_i, C, τ): By using this oracle, the adversary $\mathcal{A}$ can obtain the signature σ on C under the group private key gsk_i corresponding to the epoch τ for the honest device RD_i. Then, the oracle adds the signature σ to the SL list.

$\mathrm{Ch}_b(info_\tau, RD_{i_0}, RD_{i_1}, C^*)$: Given two devices $\mathrm{RD}^*_{i_0}$, $\mathrm{RD}^*_{i_1}$ chosen by the adversary $\mathcal{A}$ and a ciphertext C^*, the oracle outputs the challenge signature σ^* on C^* under $\mathrm{RD}^*_{i_b}$, where $b \in \{0, 1\}$ is selected randomly by the oracle.

Trace$(C, \sigma, info_\tau)$: Using this oracle, $\mathcal{A}$ can obtain the result of running the Trace algorithm on the ciphertext C, the signature σ, and $info_\tau$. The challenge signature output by Ch_b is not allowed to be queried using the oracle.

Additionally, we define the auxiliary function:

IsActive$(reg, info_\tau, RD_i)$: By querying reg and $info_\tau$, this function determines whether RD_i is in an active state. If it is active in epoch τ, outputs 1; otherwise, outputs 0.

Definition 3 (Correctness): For any PPT adversary $\mathcal{A}$, the experiment $Exp^{\mathrm{corr}}_{\mathcal{A}}(\theta)$ is defined in Fig. 3. If the advantage

$$Adv^{\mathrm{corr}}_{\mathcal{A}}(\theta) = Pr[Exp^{\mathrm{corr}}_{\mathcal{A}}(\theta) = 1] \tag{2}$$

is negligible for the adversary $\mathcal{A}$, then the scheme satisfies the correctness.

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Experiment: $\mathrm{Exp}_{A}^{\mathrm{corr}}(\theta)$

1) $(gpk, esk, tsk, reg, info) \leftarrow ENGen(\theta); HDL := \emptyset;$

2) $A \rightarrow AddD(\cdot), RReg(\cdot), GUpdate(\cdot);$

3) $(RD_i, C, \tau) \leftarrow A(gpk, info);$

4) Return 0 *if* $RD_i \in HDL$ *or* $gsk_i = \perp$ *or* $\mathrm{IsActive}(info_\tau, reg, RD_i);$

5) $\sigma \leftarrow Sign(gpk, gsk_i, info_\tau, C)$ and (RD_i^*, π_{tr}) $\leftarrow Trace(gpk, tsk, info_\tau, reg, C, \sigma);$

6) Return 1 *if* $\mathrm{Verify}(gpk, info_\tau, C, \sigma) \rightarrow 0$ *or* $RD_i \neq RD_i^*$ $Judge(gpk, RD_i, info_\tau, \pi_{tr}, C, \sigma) \rightarrow 0.$

Fig. 3. Definition of Correctness Experiment.

Experiment: $\mathrm{Exp}_{A}^{\mathrm{an}}(\theta)$

1) $(gpk, esk, tsk, info) \leftarrow ENGen(\theta);\ HDL, CDL, BDL, SL := \emptyset;$

2) $A \rightarrow AddD(\cdot), CrptD(\cdot), Reveal(\cdot), SndToD(\cdot), Trace(\cdot), WReg(\cdot), Ch_b(\cdot);$

3) $b^* \leftarrow A(gpk, esk, st);$

4) Return 1 *if* b = b^*, *and* 0 *otherwise*.

Fig. 4. Definition of Anonymity experiment.

Experiment: $\mathrm{Exp}_{A}^{\mathrm{nf}}(\theta)$

1) $(gpk, esk, tsk, info) \leftarrow ENGen(\theta); HDL, CDL, BDL, SL := \emptyset;$

2) $A \rightarrow CrptD(\cdot), Reveal(\cdot), SndToD(\cdot), WReg(\cdot), Sign(\cdot);$

3) $(\sigma, C, RD_i, \pi_{tr}, info_\tau) \leftarrow A(gpk, esk);$

4) Return 1 *if* (i) $\wedge$ (ii) $\wedge$ (iii) holds :

i) $\mathrm{Verify}(gpk, info_\tau, C, \sigma) = 1,$

ii) $Judge(gpk, RD_i, info_\tau, \pi_{tr}, C, \sigma) = 1,$

iii) $RD_i \in HDL \backslash BDL \wedge RD_i \notin CDL \wedge (C, \sigma, \tau) \notin SL;$

5) O*therwise return* 0.

Fig. 5. Definition of Non-Frameability experiment.

Experiment: $\mathrm{Exp}_{A}^{\mathrm{trace}}(\theta)$

1) $(gpk, esk, tsk, info) \leftarrow ENGen(\theta); HDL, CDL, BDL, SL := \emptyset;$

2) $A \rightarrow AddD(\cdot), CrptD(\cdot), SndToEN(\cdot), Reveal(\cdot), RReg(\cdot), Sign(\cdot), GUpdate(\cdot);$

3) $(\sigma, C, \tau) \leftarrow A(gpk, info);$

4) *if* $\mathrm{Verify}(gpk, info_\tau, C, \sigma) = 0$, *return* 0 :

5) $(RD_i, \pi_{tr}) \leftarrow Trace(gpk, tsk, info_\tau, reg, C, \sigma);$

6) Return 1 *if* (i) $\vee$ (ii) $\vee$ (iii) holds :

i) $\mathrm{IsActive}(info_\tau, reg, RD_i) = \perp,$

ii) $Judge(gpk, RD_i, info_\tau, \pi_{tr}, C, \sigma) = 0,$

iii) $RD_i = 0;$

7) O*therwise return* 0.

Fig. 6. Definition of Traceability experiment.

Definition 4 (Anonymity): For any PPT adversary $\mathcal{A}$, the experiment $Exp_{\mathcal{A}}^{\mathrm{an}}(\theta)$ is defined in Fig. 4. If the advantage

$$Adv_{\mathcal{A}}^{\mathrm{an}}(\theta) = \left| Pr[Exp_{\mathcal{A}}^{\mathrm{an}}(\theta) = 1] - \frac{1}{2} \right| \tag{3}$$

is negligible for the adversary $\mathcal{A}$, then the scheme satisfies anonymity.

Definition 5 (Non-Frameability): For any PPT adversary $\mathcal{A}$, the experiment $Exp_{\mathcal{A}}^{\mathrm{nf}}(\theta)$ is defined in Fig. 5. If the advantage

$$Adv_{\mathcal{A}}^{\mathrm{nf}}(\theta) = Pr[Exp_{\mathcal{A}}^{\mathrm{nf}}(\theta) = 1] \tag{4}$$

is negligible for the adversary $\mathcal{A}$, then the scheme satisfies non-frameability.

Definition 6 (Traceability): For any PPT adversary $\mathcal{A}$, the experiment $Exp_{\mathcal{A}}^{\mathrm{trace}}(\theta)$ is defined in Fig. 6. If the advantage

$$Adv_{\mathcal{A}}^{\mathrm{trace}}(\theta) = Pr[Exp_{\mathcal{A}}^{\mathrm{trace}}(\theta) = 1] \tag{5}$$

is negligible for the adversary $\mathcal{A}$, then the scheme satisfies traceability.

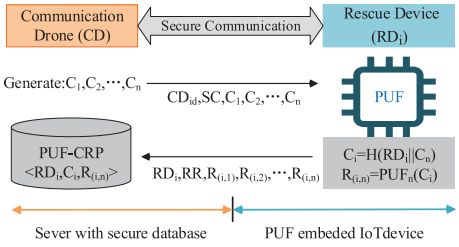


Fig. 7. Registration of Rescue Device.

V. Proposed Scheme

In this section, we first outline the device authentication process, explain the dynamic authentication strategy, and finally describe the group signature algorithm in detail.

A. System Initialization

In this phase, CS initializes the system parameters. The CS selects an elliptic curve E over $\mathbb{F}_p$, constructs a cyclic group G of order q with P as its generator, and randomly chooses $x \in \mathbb{Z}_q^*$ as the private key and computes the public key $X_{\mathrm{CS}} = x \cdot P$. Finally, the CS selects a hash function $H : \{0,1\}^* \rightarrow \mathbb{Z}_q^*$ and publishes the public parameters $\{q, G, E, P, X_{\mathrm{CS}}, H\}$.

B. CD Registration Phase

The process of CD registration to CS is as follows:

1) CD→ CS: CD sends a registration request to the CS, including the identity CD_{id}, a timestamp T_0, and the encrypted SCD_{id} using their shared key SSC.
2) CS →CD: The CS checks the validity of SCD_{id}, then verifies that $|T_{\mathrm{current}} - T_0| \leq \Delta T$. It calculates $X_{CD} = Enc_{SSC}(H(CD_{id} \,\|\, X_{CS} \,\|\, T_0))$ as the CD's authentication credential, where X_{CS} is the public key of the CS. Finally, the CS sends $\{CD_{\mathrm{id}}, X_{\mathrm{CD}}, T_0\}$ to the CD and adds CD_{id} to the registration list RL.
3) CD: Upon receiving the message, the CD stores X_{CD}.

C. Device Registration Phase

The device RD_i registers with the CD in a secure environment. The detailed process is shown in Fig. 7.

1) CD → RD_i: CD sends a set of challenges $\{C_1, \ldots, C_n\}$, its own identity CD_{id}, and the encrypted SC using the shared secret key SRC between them.
2) RD_i → CD: RD_i verifies the validity of SC, calculates the corresponding responses using PUF_i functions (where $i = 1, 2, \ldots, n$), and encrypts them using the shared secret key SRC to obtain RR.

$$R_{(i,n)} = PUF_n(C_i) = PUF(H(RD_i \| C_n)). \tag{6}$$

RD_i then sends $\{RD_i, R_{(i,1)}, \ldots, R_{(i,n)}, RR\}$ to CD.
3) CD: For each RD_i, CD possesses n different CRP pairs $(C_i, R_{(i,n)})$, and stores them in a secure database.

D. CD Authentication Phase

After registration, if CD requests resources from CS, authentication is required. The process is as follows:

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1) CD→ CS: CD sends an request to CS, including its identity CD_{id}, a random number r_k, a timestamp T_1, and the encrypted CR using their shared key SSC.
2) CS →CD: CS checks the RL to confirm whether CD_{id} is registered and verifies the legitimacy of the CR. If established, CS combines the timestamp T_1 and its private key x to generate a credential:

$$Y_{CD} = H(CD_{id} \| r_k \| X_{CS} \| T_1). \tag{7}$$

CS then sends $\{T_1, Y_{CD}\}$ to CD.
3) CD → CS: CD verifies CS's identity by recalculating:

$$Y^*_{CD} = H(CD_{id} \| r_k \| X_{CS} \| T_1). \tag{8}$$

If $Y^*_{CD} = Y_{CD}$, CS's authentication is successful. Meanwhile, CD generates random number r_{k1}, calculates Z_{CD} and SC_2, and sends them to CS.

$$Z_{CD} = H(CD_{id} \| r_{k1} \| X_{CS} \| T_1 \| r_k). \tag{9}$$

$$SC_2 = Enc_{SSC}(Z_{CD} \| r_{k1}) \tag{10}$$

4) CS: CS verifies CD's identity by recalculating Z^*_{CD} and comparing it with the received Z_{CD}. If the two match, CD's authentication is successful.

Algorithm 1 Dynamic Authentication Strategy

Input: PUF response sequence $R^*_{(i,1)}$, $R^*_{(i,2)}$, $R^*_{(i,3)}$ received from the device end.
Output: 0 or 1.
Case 1: In extremely harsh environments with low safety requirements
if $R^*_{(i,1)} == R_{(i,1)}$ or $R^*_{(i,2)} == R_{(i,2)}$ or $R^*_{(i,3)} == R_{(i,3)}$ **then**
    return 1
else
    return 0
end if
Case 2: In general environments with low safety requirements
if ($R^*_{(i,1)} == R_{(i,1)}$ and $R^*_{(i,2)} == R_{(i,2)}$) or ($R^*_{(i,1)} == R_{(i,1)}$ and $R^*_{(i,3)} == R_{(i,3)}$) or ($R^*_{(i,2)} == R_{(i,2)}$ and $R^*_{(i,3)} == R_{(i,3)}$) **then**
    return 1
else
    return 0
end if
Case 3: In an environment with extremely high security requirements
if $R^*_{(i,1)} == R_{(i,1)}$ and $R^*_{(i,2)} == R_{(i,2)}$ and $R^*_{(i,3)} == R_{(i,3)}$ **then**
    return 1
else
    return 0
end if

E. Device Authentication Phase

At this stage, a PUF-based multi-factor authentication circuit was constructed, as shown in Fig. 9. By leveraging its physical characteristics, a PUF-based dynamic authentication strategy is designed to achieve the dynamic authentication of RD_i under different security requirements, as shown in Algorithm 1.

Assume that each device RD_i is embedded with n PUFs, as shown in Fig. 8. In our scheme, taking $n = 3$ as an example, the embedded PUFs are Arbiter PUF, Ring Oscillator PUF, and SRAM PUF, as shown in Fig. 9. The process is as follows:

1) CD → RD_i: CD sends a set of challenges $\{C_1, C_2, C_3\}$, its own identity CD_{id}, and the encrypts them using the shared secret key SRC to obtain SC.

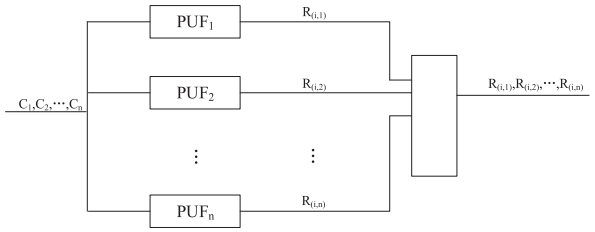


Fig. 8. Schematic diagram of n PUFs.

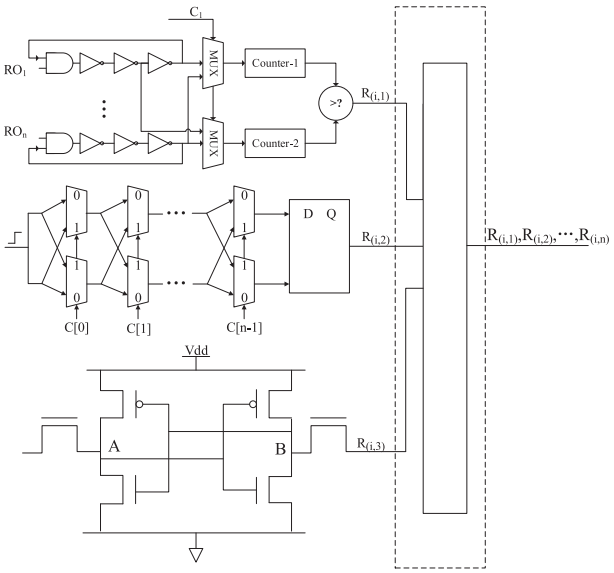


Fig. 9. Multi-factor puf authentication circuit.

Generate common parameter values	
The trusted party selects and publishes public parameters (n, q) and polynomial $f \in R_q$, q is a positive integer.	
Alice	*Bob*
Key Generation	
Select $s \in$ as the private key, and select $e \in R$ from χ; Calculate the public key pair (f, l) as $l = f \cdot s + e$.	
$(f\ l)$ →	
	Encryption
	Select plaintext $p \in \{0,1\}^n$; Select $t \in R_q$ and select $e_1, e_2 \in R$; Calculate the ciphertext using (f, l): $c_1 = f \cdot t + e_1 \ (mod\ q)$, $c_2 = l \cdot t + e_2 + \lfloor q/2 \rceil \cdot p \ (mod\ q)$; Send ciphertext c_1 and c_2 to Alice.
← c_1 and c_2	
Decryption	
Calculate $p = c_2 - c_1 \cdot s \ (mod\ q)$.	

Fig. 10. Lattice-based encryption algorithm.

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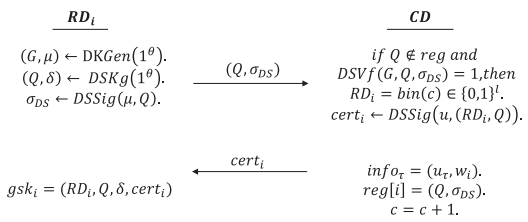


Fig. 11. Overview of the < Join, Issue> protocol.

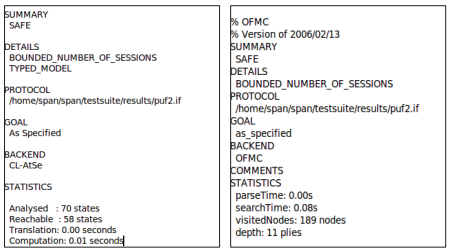


Fig. 12. Security verification results using AVISPA's CL-AtSe and OFMC.

2) $RD_i \rightarrow CD$: RD_i verifies the validity of SC, and uses the PUF functions to compute the responses:

$$R^*_{(i,1)} = PUF_1(C_i) = PUF_1(H(RD_i \| C_1)), \quad (11)$$
$$R^*_{(i,2)} = PUF_2(C_i) = PUF_2(H(RD_i \| C_2)), \quad (12)$$
$$R^*_{(i,3)} = PUF_3(C_i) = PUF_3(H(RD_i \| C_3)), \quad (13)$$
$$RC = Enc_{SRC}(RD_i \| R^*_{(i,1)} \| R^*_{(i,2)} \| R^*_{(i,3)}). \quad (14)$$

RD_i sends$\{RD_i, R^*_{(i,1)}, R^*_{(i,2)}, R^*_{(i,3)}, RC\}$ to CD.

3) CD: CD verifies the validity of RC, and dynamically authenticates the device based on the strategy. The process is as follows:

a) In Case 1, authentication succeeds if any condition is satisfied; otherwise, it fails.

$$R^*_{(i,1)} == R_{(i,1)} \ or \ R^*_{(i,2)} == R_{(i,2)} \ or \ R^*_{(i,3)} == R_{(i,3)}. \quad (15)$$

b) In Case 2, authentication succeeds if any set of the following conditions is satisfied; otherwise, it fails.

$$(R^*_{(i,1)} == R_{(i,1)} \ and \ R^*_{(i,2)} == R_{(i,2)}) \ or \ (R^*_{(i,1)} == R_{(i,1)} \ and \ R^*_{(i,3)} == R_{(i,3)}) \ or \ (R^*_{(i,2)} == R_{(i,2)} \ and \ R^*_{(i,3)} == R_{(i,3)}). \quad (16)$$

c) In Case 3, authentication succeeds if all of the following conditions are satisfied; otherwise, it fails.

$$R^*_{(i,1)} == R_{(i,1)} \ and \ R^*_{(i,2)} == R_{(i,2)} \ and \ R^*_{(i,3)} == R_{(i,3)}. \quad (17)$$

F. Group Signature

The scheme uses group signature to anonymously protect device identities and prevent privacy leakage. Meanwhile, a lattice-based encryption algorithm is introduced for message transmission to resist quantum computing attacks and ensure message integrity. The algorithm is shown in Fig. 10.

1) Initialization: The trusted party randomly selects a polynomial $f \in R_q$ that is shared among all devices.
2) Key Generation: Alice randomly selects a polynomial $s \in R_q$, and error polynomial $e \in R$, and computes the public key:

$$l = f \cdot s + e. \quad (18)$$

3) Encryption: For an n-bit plaintext $p \in \{0,1\}^n$, Bob randomly selects a polynomial $t \in R_q$ and error polynomials $e_1, e_2 \in R$, and computes ciphertext $C = (c_1, c_2)$:

$$c_1 = f \cdot t + e_1 \ (mod\, q),$$
$$c_2 = l \cdot t + e_2 + \left\lfloor \frac{q}{2} \right\rfloor \cdot p \ (mod\, q). \quad (19)$$

4) Decryption: Alice can recover the original message p by calculating the equation 20.

$$c_2 - c_1 \cdot s = (t \cdot e - s \cdot e_1 + e_2) + \left\lfloor \frac{q}{2} \right\rfloor \cdot p \ (mod\, q). \quad (20)$$

Next, we provide a detailed explanation of the group signature process.

Algorithm 2 ENGen

Input: θ, N
Output: $pp, gpk, esk, tsk, info$
Choose $H_1 : \{0,1\}^* \rightarrow \{1,2,3\}^k$, where $k = \omega(\log \theta)$
Choose $H_2 : \{0,1\}^* \rightarrow \mathbb{Z}^*_q$
Choose $u \leftarrow \mathbb{Z}^*_q$, set $U = u \cdot G$
Choose $s_1, s_2 \in \mathbb{R}_q$, $e_1, e_2 \in \mathbb{R}$
Compute $l_1 = f_1 \cdot s_1 + e_1$ (TM's public key)
Compute $l_2 = f_2 \cdot s_2 + e_2$ (receiver's public key)
Initialize $info = \emptyset$, $T = \emptyset$, $reg = \emptyset$, $c = 0$
Return: $pp = (q, G, E, P, H, H_1, N, d)$, $gpk = (pp, U, f_1, l_1)$, $esk = u$, $tsk = (s_1, e_1)$, $info = \emptyset$.

ENGen($1^\theta, N$): Enter security parameter 1^ϑ and group size N output the public parameters pp, the group public key gpk, CD's private key esk, TM's private key tsk, and group information $info$. The process is shown in Algorithm 2.

DKGen(1^θ): The device RD_i randomly selects $\mu \in \mathbb{Z}^*_q$ and computes $G = \mu \cdot P$. Its public-private key pair as $(dpk_i, dsk_i) = (G, \mu)$.

Join($info_\tau, gpk, dsk_i$) $\leftrightarrow$ Issue($info_\tau, esk, dpk_i$): At epoch τ, new members joins the group, as shown in Fig. 11. The process is as follows:

1) RD_i generates a key pair $(pk_i, sk_i) = (Q, \delta)$ using DSKg(1^θ), signs with DSSig(μ, Q) to produce a signature σ_{DS}, and sends(Q, σ_{DS}) to CD.
2) If the following conditions are met: $Q \notin$ reg and DSVf$(G, Q, \sigma_{DS}) = 1$, CD will generate a certificate $cert_i \leftarrow$ DSSig$(u, (RD_i, Q))$ and send it to RD_i.
3) Finally, RD_i holds the group key gsk_i. CD updates the group information info$_\tau$, saves (Q, σ_{DS}) in the reg.

GUpdate($gpk, esk, info_\tau, S, reg$): During epoch τ, group members are revoked using this algorithm. The process is as follows:

1) First, let S contain the public keys of the members to be revoked. If $S = \emptyset$, return $\perp$; otherwise, for $i \in [0, N-1]$, $S = \{reg[1], reg[2], \ldots, reg[i]\}$.
2) CD searches the tree T to locate and replace the public keys of the revoked members with zero values, updating the witness information $info_{\tau_{new}}$ and the list reg. In other words, the non-zero public keys $\{Q_j\}_j$ of active

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members are accumulated into the root node $u_{\tau_{new}}$, where the witness information w_j is set for each j.

3) Finally, CD publishes the updated group information $info_{\tau_{new}} = (u_{\tau_{new}}, \{w_j\}_j)$.

Sign($gpk, gsk_i, info_\tau, C$): The group member uses the algorithm in Fig. 10 to encrypt the message, ensuring its integrity. Then, using Algorithm 3, the group member signs the ciphertext C with the group key gsk_i at epoch τ.

1) First, The device RD_i checks whether $info_\tau$ contains its own witness information. If not, return $\perp$. Otherwise, the RD_i retrieves the root value u_τ.
2) Using the symmetric key K, the receiver's public key (f_2, l_2), and TM's public key (f_1, l_1) to encrypt the message m, K, and d, respectively, where $d = H(K\|RD_i)$. The resulting ciphertext is $C = (C_s, C_r, C_t)$.
3) The RD_i computes the hash of the ciphertext $z = H(C) \in \mathbb{Z}_q^*$, and randomly selects $k \in \mathbb{Z}_q^*$ such that $k \neq 0$, and calculates:

$$r = (k \cdot P).x \bmod q, \tag{21}$$

$$p = k^{-1} \cdot (z + \delta \cdot r) \bmod q. \tag{22}$$

where $k \cdot P$ is a point on E, and its x-coordinate is taken.

Algorithm 3 Sign

Input: $gpk, gsk_i, \text{info}_\tau, C$.
Output: σ.
if w_i in info_τ **then**
  return u_τ.
else
  return $\perp$.
end if
Compute $d = H(K \parallel RD_i)$
Choose $t_1, t_2 \leftarrow \chi, e_{1,1}, e_{1,2} \leftarrow \chi^l, e_{2,1}, e_{2,2} \leftarrow \chi^l$
Compute $C = (C_s, C_r, C_t)$
$C_s = \text{Enc}(K, m)$
$C_t = (f_1 \cdot t_1 + e_{1,1}, l_1 \cdot t_1 + e_{1,2} + \lfloor q/2 \rfloor \cdot d)$
$C_r = (f_2 \cdot t_2 + e_{2,1}, l_2 \cdot t_2 + e_{2,2} + \lfloor q/2 \rfloor \cdot K)$
Calculate the hash value of C: $z = H(C) \in \mathbb{Z}_q^*$
Choose $k \leftarrow \mathbb{Z}_q^*$ and $k \neq 0$
Compute $r = (k \cdot P).x \mod q$
Compute $p = k^{-1} \cdot (z + \delta \cdot r) \mod q$
return $\sigma = (r, p, u_\tau)$

4) Finally, output the signature $\sigma = (r, p, u_\tau)$.

Verify($gpk, info_\tau, C, \sigma$): The verifier uses this algorithm to verify the signature σ. The process is as follows:

1) Retrieving the current root value u_τ^* from the group information $info_\tau$.
2) The verifier computes the hash value of the ciphertext $z^* = H(C) \in \mathbb{Z}_q^*$, and calculates:

$$u_1 = p^{-1} \cdot z^* \bmod q, \tag{23}$$

$$u_2 = p^{-1} \cdot r \bmod q, \tag{24}$$

$$T = u_1 \cdot P + u_2 \cdot Q \bmod q. \tag{25}$$

$$v = T.x \bmod q. \tag{26}$$

3) if $v = r$ and $u_\tau^* = u_\tau$, output 1; otherwise, output 0.

Trace($gpk, tsk, info_\tau, C, \sigma, reg$): The TM uses this algorithm to trace malicious group members.

1) If Verify $(gpk, info_\tau, C, \sigma) = 0$, abort. Otherwise, the TM uses its private key $tsk = (s_1, e_1)$ to decrypt C_t and obtain d to recover the symmetric key K and the identity RD_i^*.

$$d = \left\lfloor \frac{c_{t_2} - c_{t_1} \cdot s_1}{\lfloor q/2 \rfloor} \right\rfloor \tag{27}$$

2) The TM checks whether $info_\tau$ contains the witness information of RD_i^*. If it does and $reg[RD_i^*] \neq \emptyset$, generate the proof $\pi_{tr} = (Q, \sigma_{DS})$, and return the identity RD_i^* and π_{tr}. Otherwise, return $\perp$.

Judge($gpk, RD_i, info_\tau, C, \sigma, \pi_{tr}$): Verify the validity of the trace proof π_{tr} output by the Trace algorithm. If valid, output 1; otherwise, output 0.

VI. Security Analysis

This section verifies the security of the scheme through the Random Oracle Model (ROM) and informal analysis, and performs formal verification using the AVISPA tool.

A. Analysis Under the Random Oracle Model

The PUF-CLGS satisfies correctness, anonymity, non-frameability, and traceability.

Theorem 1: The PUF-CLGS scheme satisfies correctness.

Proof: The correctness of the PUF-CLGS scheme relies on the correctness of the Sign, Verify, Trace, and Judge algorithms. $\mathcal{A}$ valid signature $\sigma = (r, p, u_\tau)$ can be verified by any group member using Verify. If $v = r$ and $u_\tau^* = u_\tau$, the verification is successful. Next, the Trace algorithm calculates d^* as follows:

$$\begin{aligned} d^* &= c_{t_2} - c_{t_1} \cdot s_1, \\ &= l_1 \cdot t_1 + e_{1,2} + \left\lfloor \frac{q}{2} \right\rfloor \cdot d - (f_1 \cdot t_1 + e_{1,1}) \cdot s_1, \\ &= e_1 \cdot t_1 + e_{1,2} + \left\lfloor \frac{q}{2} \right\rfloor \cdot d - e_{1,1} \cdot s_1. \end{aligned} \tag{28}$$

And recovers d from d^* based on the following rule: if d^* is close to 0, then $d^* = 0$; otherwise, $d^* = 1$. The Trace algorithm outputs d with overwhelming probability and retrieves the signer identity RD_i^*. Since the member is active, $info_\tau$ contains $(RD_i^*, w_\ell, \ldots, w_1)$, and $reg[RD_i^*] \neq \emptyset$, proving the correctness of the Trace algorithm. Finally, the Judge algorithm verifies the proof π_{tr} with probability 1.

Theorem 2: The PUF-CLGS scheme satisfies anonymity in the Random Oracle Model (ROM).

Proof: Assume that adversary $\mathcal{A}$ interacts with challenger $\mathcal{C}$, and $\mathcal{C}$ simulates $\mathcal{A}$'s attack through query operations and attempts to guess $\mathcal{A}$'s choice. The goal of $\mathcal{C}$ is to guess the challenge b in the anonymity game. The process is as follows:

1) Setup: Let P be the generator of the elliptic curve. $\mathcal{C}$ does the following: First, $\mathcal{C}$ randomly selects $u \in \mathbb{Z}_q^*$, computes $U = u \cdot P$. $\mathcal{C}$ selects $s_1 \in R_q$ and $e_1 \in R$, and computes the public key $l_1 = f_1 \cdot s_1 + e_1$. $\mathcal{C}$ generates the group public key $gpk = (pp, U, f_1, l_1)$ and sends it to $\mathcal{A}$. Then, it randomly selects two devices RD_{i_0} and RD_{i_1}, secretly holds their identities, and generates key pairs (G_0, μ_0) and (G_1, μ_1).
2) Hash queries: $\mathcal{A}$ can query hash oracles H_1 and H.

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3) Phase 1: $\mathcal{A}$ can query the oracles in Fig. 4. The processes follow the definitions in the Security Model. $\mathcal{C}$ responds as follows: For $RD_i \in \{RD_{i_0}, RD_{i_1}\}$, $\mathcal{C}$ constructs their signature. $\mathcal{C}$ encrypts the message m and generates the ciphertext $C = (C_s, C_r, C_t)$. It selects $k_0 \in \mathbb{Z}_q^*$ and computes $info_{\tau_0} = (u_{\tau_0}, w_{i_0})$, obtaining the signature $\sigma_0 = (r_0, p_0, u_{\tau_0})$, where $r_0 = (k_0 \cdot G).x \bmod q$, $p_0 = k_0^{-1} \cdot (H(C) + \delta \cdot r_0) \bmod q$. Similarly, the signature for device RD_{i_1} is $\sigma_1 = (r_1, p_1, u_{\tau_1})$. Finally, If $\mathcal{A}$ uses Oracle CrptD to query devices RD_{i_0} or RD_{i_1}, $\mathcal{C}$ aborts.
4) Challenge: $\mathcal{A}$ selects two devices $RD_{i_0}^*, RD_{i_1}^*$. If $(RD_{i_0}^*, RD_{i_1}^*) \neq (RD_{i_0}, RD_{i_1})$, then $\mathcal{C}$ aborts. Otherwise, $\mathcal{C}$ randomly selects $b \in \{0, 1\}$, generates the signature $\sigma^* = (r^*, p^*, u_\tau^*)$, and sends it to $\mathcal{A}$.
5) Phase 2: Same as Phase 1, except that $\mathcal{A}$ cannot query Oracle CrptD for RD_{i_0} and RD_{i_1}.
6) Output: $\mathcal{A}$ outputs $b^* \in \{0, 1\}$. If $b^* = b$, $\mathcal{C}$ succeeds; otherwise, $\mathcal{C}$ fails.

Assume that the probability of $\mathcal{A}$ successfully guessing b is $\Pr[b = b^*] > \frac{1}{2} + \epsilon$. When $b^* = b$, $\Pr[b = b^*] = \frac{1}{2}$. $\mathcal{C}$ has at least $\frac{\epsilon}{2}$ advantage in breaking the anonymity. $\mathcal{C}$ terminates with a probability of at most $\frac{q_j}{q}$ by querying the Oracle SndToD, where q_j is the number of queries. Hence, $\mathcal{C}$'s probability of termination due to $\mathcal{A}$'s anonymity attack is at most $\frac{q_P q_j}{q}$. Since $\mathcal{A}$ cannot gain any information about $RD_{i_0}^*$ and $RD_{i_1}^*$, the probability of termination caused by $\mathcal{A}$'s challenge or query is at most $\frac{1}{n^2}$. Thus, $\mathcal{C}$'s advantage in breaking the anonymity is at least:

$$\frac{\epsilon}{2}\left(\frac{1}{n^2} - \frac{q_G q_j}{q}\right). \tag{29}$$

Theorem 3: If the ECDLP problem holds on G, and the underlying encryption scheme is based on the RLWE problem, PUF-CLGS satisfies non-frameability in the ROM.

Proof: Assume that adversary $\mathcal{A}$ can break the non-frameability of PUF-CLGS. We construct a challenger $\mathcal{C}$ that breaks the ECDLP problem. Given a tuple ($P^* = a \cdot P$, $P^* \in G$, $a \in \mathbb{Z}_q^*$), $\mathcal{C}$ needs to compute a. $\mathcal{C}$ randomly selects $u \in \mathbb{Z}_q^*$, computes $U = u \cdot G$. Randomly selects $s_1 \in R_q$ and $e_1 \in R$, computes $l_1 = f_1 \cdot s_1 + e_1$, and sends gpk and esk to $\mathcal{A}$. $\mathcal{C}$ simulates devices $\{RD_{i_1}, \ldots, RD_{i_q}\}$, and selects RD_{i_j} from them, and provides the following oracles to $\mathcal{A}$:

1) SndToD(RD_i): If $RD_i \neq RD_{i_j}$, $\mathcal{C}$ simulates an honest device RD_i. Otherwise, $\mathcal{C}$ uses $G_{i_j} = G^*$ to run the $\langle$Join, Issue$\rangle$ protocol. The group private key for RD_{i_j} is $gsk_{i_j} = (RD_{i_j}, Q_{i_j}, \delta_{i_j}, cert_{i_j})$, where $\mathcal{C}$ does not know $\delta_{i_j} = a$.
2) $\mathcal{C}$ can simulate the following oracles: Oracle WReg, Oracle Sign, Oracle Reveal, Oracle CrptD, and Oracle SndToD, except when Oracle Reveal(RD_{i_j}) is queried.

Assume that $\mathcal{A}$ succeeds with probability ϵ, then the probability of outputting a valid pair (C, σ) is at least ϵq, where $RD_{i_j} \in \{RD_{i_1}, \ldots, RD_{i_q}\}$. Using the general forking lemma, there exists an adversary $\mathcal{A}^*$ that can output two valid signatures $\sigma = (r, p, u_\tau)$ and $\sigma^* = (r^*, p^*, u_\tau^*)$, satisfying $p \neq p^*$. Through the correctness proof of Theorem 1, we can find $z = H(C) \in \mathbb{Z}_q^*$, such that: $T = p^{-1} \cdot z \cdot P + p^{-1} \cdot r \cdot Q \bmod q$, $v = T.x \bmod q$, where:

$$C = (C_s, C_r, C_t),\ C_s = \mathrm{Enc}(m, K),$$
$$C_r = (f_2 \cdot t_2 + e_{2,1}, l_2 \cdot t_2 + e_{2,2} + \lfloor q/2 \rfloor \cdot K),$$
$$C_t = (f_1 \cdot t_1 + e_{1,1}, l_1 \cdot t_1 + e_{1,2} + \lfloor q/2 \rfloor \cdot d). \tag{30}$$

Since the underlying encryption scheme is based on the RLWE problem, the non-frameability of PUF-CLGS is guaranteed.

Theorem 4: The PUF-CLGS scheme satisfies traceability under ROM.

Proof: If there is no polynomial-time adversary capable of forging a valid signature that is verifiable but cannot be traced, then PUF-CLGS is traceable. We consider the game between the adversary $\mathcal{A}$ and the challenger $\mathcal{C}$ as follows.

1) Setup: $\mathcal{C}$ constructs an elliptic curve instance E, sets P as the generator. $\mathcal{C}$ randomly selects $u \in \mathbb{Z}_q^*$, computes $U = u \cdot P$. And randomly selects $s_1 \in R_q$, $e_1 \in R$, computes the public key $l_1 = f_1 \cdot s_1 + e_1$. Finally, $\mathcal{C}$ sends $gpk = (pp, U, f_1, l_1)$ and $tsk = (s_1, e_1)$ to $\mathcal{A}$.
2) Hash queries: $\mathcal{A}$ can query the hash oracle at any time.
3) Queries Phase: $\mathcal{A}$ can query the oracles in Fig. 6. $\mathcal{C}$ responds as follows: Constructs the signature of RD_i. $\mathcal{C}$ first encrypts the message m and generates ciphertext $C = (C_s, C_r, C_t)$. Next, $\mathcal{C}$ uses the group private key of device RD_i, $gsk_i = (RD_i, Q, \delta, cert)$, to generate the signature $\sigma = (r, p, u_\tau)$ on the ciphertext C.
4) Forge: $\mathcal{A}$ outputs a signature σ on the ciphertext C.

According to the definition of the traceability game, if $\mathcal{A}$ wins the game at epoch τ, then $\mathrm{Verify}(gpk, \mathrm{info}_\tau, C^*, \sigma^*) = 1$ holds, and one of the following conditions is satisfied.

1) $\mathrm{Trace}(gpk, tsk, \mathrm{info}_\tau, C, \sigma, \mathrm{reg}) = \perp$: the Trace algorithm cannot find a valid device RD_i that satisfies the signature. Since $C_t = (f_1 \cdot t_1 + e_{1,1}, l_1 \cdot t_1 + e_{1,2} + \lfloor q/2 \rfloor \cdot d)$ and the encryption scheme is based on the RLWE problem, $\mathcal{A}$ cannot modify d and the reg, and thus cannot use non-member public key to generate signature.
2) $\mathrm{Trace}(gpk, tsk, \mathrm{info}_\tau, C^*, \sigma^*, \mathrm{reg}) = (i^*, \pi_{tr}^*) \wedge \mathrm{Judge}(gpk, i^*, \mathrm{info}_\tau, C^*, \sigma^*, \pi_{tr}^*) = 0$: the Trace algorithm returns device RD^* and proof π_{tr}^*, but the Judge algorithm outputs 0. $\mathcal{A}$ uses d^* of RD_i^* to generate signature σ^*. If d^* in σ^* is decrypted by C_t and corresponds to a group member, then $d = d^*$. If $\mathcal{A}$ attempts to make $d \neq d^*$, it contradicts the unforgeability of the underlying signature scheme.

B. Formal Security Verification Using AVISPA Tools

AVISPA (Automated Validation of Internet Security Protocols and Applications) is a verification tool that supports the use of High-Level Protocol Specification Language (HLPSL) to describe protocols and their security properties [34]. The proposed scheme adopts the DY attacker model, which allows intruders to simulate the behavior of legitimate participants. The registration and authentication phases are coded using HLPSL and simulated using the SPAN tool, and formally verified using the On-the-Fly Model Checker (OFMC) and Constraint-Logic-based Attack Searcher (CL-AtSe) module [35]. The simulation results are shown in Fig. 12. The proposed scheme can resist replay attacks, MITM attacks and other attacks. The HLPSL security verification program is available at https://github.com/tangruixia/PUF-CLGS.

C. Informal Analysis

1) Unforgeability: The security of the PUF-CLGS scheme is based on the RLWE and ECDLP problems, which are

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TABLE III
COMPARISON OF SECURITY PROPERTIES

Requirement	[17]	[19]	[20]	[21]	[22]	[25]	Ours
Correctness	✓	✓	✓	✓	✓	✓	✓
Anonymity	✓	✓	✓	✓	✓	✓	✓
Traceability	✓	✓	✓	✓	✓	✓	✓
Non-frameability	✓	✓	✓	✓	✓	✓	✓
Unforgeability	✓	✓	✓	✓	✓	✓	✓
Eavesdropping Attacks	×	×	✓	✓	✓	✓	✓
Quantum Computing Attacks	×	×	×	×	×	✓	✓
Message Integrity	✓	✓	✓	✓	✓	✓	✓
MITM Attacks	✓	✓	✓	✓	✓	✓	✓
Physical Clone Attacks	×	×	×	×	×	×	✓
Impersonation Attacks	✓	✓	✓	✓	✓	✓	✓
Collusion Attacks	✓	×	✓	×	×	×	✓

TABLE IV
MAIN OPERATIONS

Operation	Description
T_{em}	ECC scalar multiplication
T_{ea}	ECC point addition
T_{me}	Matrix modular exponentiation
T_{ma}	Matrix modular addition
T_p	Piring operation
T_{e,G_1}	Multiplication in G_1
T_{e,G_2}	Multiplication in G_2
T_{e,G_T}	Multiplication in G_T
T_{exp}	Exponentiation in G_T
T_{enc}	Encryption
T_{dec}	Decryption
T_h	Hash operation
T_{puf}	PUF generation

difficult for attackers to solve, making it impossible to forge valid signatures. Additionally, the physical properties of PUF ensure the non-forgeability of the device. Therefore, the scheme is unforgeable.

2) Eavesdropping Attacks: Due to the security properties of the hash function and PUF, attackers cannot obtain any useful information, and the PUF-CLGS scheme can resist eavesdropping attacks.
3) Resistance Quantum Computing Attacks: PUF-CLGS relies on the hard lattice problem, particularly the RLWE problem over polynomial rings, which is known to be quantum resistant. So the scheme can resist quantum computing attacks.
4) Resistance MITM Attacks: Due to the robustness of the verification process in PUF-CLGS, any modified message will result in the signature failing verification. Therefore, it effectively prevents MITM attacks.
5) Resistance Physical Clone Attacks: The physical properties of PUF ensure that any tampering with the device will alter its behavior, preventing tampering attacks. Thus, PUF-CLGS can resist physical clone attacks.
6) Impersonation Attacks: Since the PUF of each device is unique, attackers cannot emulate its behavior. So the PUF-CLGS can prevent impersonation attacks.
7) Resistance Collusion Attacks: Each device has different witness information *w*, and if colluders use the same w, Trace algorithm can detect and prevent collusion attacks.
8) Message Integrity: PUF-CLGS uses collision-resistant hash functions and group signature to verify the integrity of communication channel information.

VII. IMPLEMENTATION AND COMPARISONS

This section evaluates PUF-CLGS in terms of security, experimental configuration and benchmark schemes, computational overhead and communication cost, and compare it with other schemes.

A. System Security Property Analysis

Table III shows the comparison results of the PUF-CLGS with existing schemes [17], [19], [20], [21], [22], and [25] in terms of security properties. Here, ✓ indicates that the scheme satisfies the security property, while × means it does not.

As shown in Table III, PUF-CLGS not only possesses basic security attributes but also demonstrates superior resistance against quantum computing and physical cloning attacks. PUF-CLGS is built based on the RLWE problem of the lattice, ensuring that its solution complexity remains exponential in a quantum computing environment, effectively resisting quantum attacks such as the Shor algorithm. In contrast, schemes [17], [19], [20], and [21] rely on traditional number-theoretic problems such as bilinear pairing, while scheme [25] resists quantum computing attacks but does not consider physical cloning attacks. Moreover, PUF-CLGS uses the randomness of PUF hardware to generate unique responses, preventing physical tampering and cloning attacks, and providing more comprehensive security protection.

B. Experimental Configuration and Benchmark Schemes

PUF-CLGS uses a cloud host with 4 core CPU, 4GB memory and 40 GB storage, running Ubuntu 22.04 system and using Rust language to implement the scheme. The computation time of bilinear pairing and elliptic curve cryptography (ECC) was tested using the MIRACL library, and the performance of lattice cryptography was evaluated using the NFL library. The experiment was repeated 1000 times and the average value was taken. The abbreviations of the main operations are shown in Table IV.

We select [17], [19], [20], [21], and [22] as benchmark schemes for performance evaluation with PUF-CLGS. Their basic ideas are as follows:

Reference [17] aims to ensure the credibility of VANETs announcement messages and vehicle privacy. After registering with the group manager, vehicles obtain signing keys and short-term certificates, and then sign the announcement messages. The receiver verifies and links the received message, and uses the reputation mechanism to assess the credibility and prevent false announcements.

Reference [19] provides efficient authentication for IoT devices. The trusted authority (TA) initializes parameters and issues keys, the data owner signs and encrypts the data, while the data user verifies and decrypts the data. The conversion server (CS) can selectively link data and support traceability to ensure privacy control.

Reference [20] aims to perform batch verification of IoT devices, reducing reliance on central servers. The device registers with the Mobile Access Point (M-AP) to obtain a certificate, employs group signature for batch authentication, and establishes a session key upon successful authentication to ensure secure communication.

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TABLE V
COMPUTATION OVERHEAD

Algorithm	Ours	[17]	[19]	[20]	[21]	[22]
ENGen	$T_{em} + 2T_{me} + 2T_{ma}$	$2T_p + 2T_{e,G_1}$	$T_{e,G_1} + T_{e,G_2}$	$2T_{em}$	$T_p + T_{e,G_T}$	$T_p + T_{e,G_1} + T_{e,G_2}$
DKGen	T_{em}	$2T_{e,G_1}$	$2T_{e,G_1}$	$2T_{e,G_1} + 2T_h$	$2T_{e,G_1}$	$2T_{e,G_1}$
Join ↔ Issue	$4T_{em} + 3T_{puf} + 3T_h$	$2T_p + 4T_{e,G_1}$	$5T_{ea} + 7T_{e,G_1} + 2T_p$	$2T_p + 2T_{em} + 2T_h$	$2T_p + 5T_{ea} + 7T_{e,G_1}$	$2T_p + 3T_{ea} + 5T_{e,G_1}$
Sign	$2T_{enc} + 2T_{em} + T_{ea} + T_h$	$2T_p + 5T_{ea} + 14T_{e,G_1} + T_h$	$15T_{ea} + 24T_{e,G_1}$	$3T_{ea} + 5T_{em} + T_h$	$3T_{e,G_1} + 10T_{e,G_T} + 3T_p + T_h$	$15T_{ea} + 24T_{e,G_1}$
Verify	$4T_{em} + T_{ea} + T_h$	$2T_p + 15T_{e,G_1} + T_h$	$8T_{ea} + 12T_{e,G_1} + 2T_p + T_h$	$4T_{ea} + 2T_{em} + T_{e,G_T} + 5T_p$	$2T_{e,G_1} + 9T_{e,G_T} + 4T_p + T_h$	$8T_{ea} + 12T_{e,G_1} + 3T_p + T_h$
Trace	$T_{dec} + T_{em} + T_{ea}$	$2T_p + T_{e,G_1}$	$6T_{ea} + 10T_{e,G_1}$	$2T_p + T_h$	$T_{e,G_1} + 2T_{e,G_T}$	$T_{ea} + 3T_{e,G_1}$

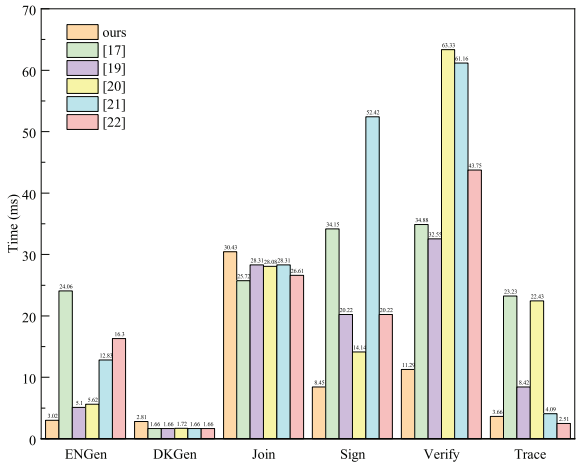


Fig. 13. Computation Overhead.

Reference [21] provides anonymous authentication for industrial IoT (IIoT). The edge intelligent controller (EIC) registers with the key server (KS) to obtain the private key and certificate, and generates anonymous authentication information, while the edge server (ES) verifies but cannot identify the device. Finally, KS can trace abnormal devices and protect the privacy of legitimate devices.

Reference [22] provides collaborative authentication for heterogeneous networks. A group signature based on the SMO algorithm is designed to protect device privacy and support traceability. Then, the consortium blockchain is employed to achieve collaborative authentication, where the access network performs preliminary verification and the core network completes global authentication.

C. Computation Overhead

The computational overhead of each phase in the PUF-CLGS is analyzed and compared with benchmark schemes [17], [19], [20], [21], and [22], with detailed results shown in Table V. As shown in Table V and Fig. 13, schemes [17], [19], [20], [21], and [22] incur significant computational overhead during the signing and verification phases due to time-consuming bilinear pairing and elliptic curve operations. In contrast, except for slightly higher overhead in the key generation and group joining phases, the PUF-CLGS exhibits significantly lower costs in other phases compared to the benchmark schemes [17], [19], [20], [21], and [22]. In particular, the computational cost is greatly reduced in the signing and verification phases. Overall, the PUF-CLGS is particularly suitable for IoT scenarios, especially in post-disaster rescue.

TABLE VI
COMMUNICATION OPERATION

Operation	byte size
$\|G_1\|$	128
$\|G_2\|$	272
$\|G_T\|$	384
$\|Z_q\|$	32
ECC point	32
$\|C\|$	404
Hash results	4

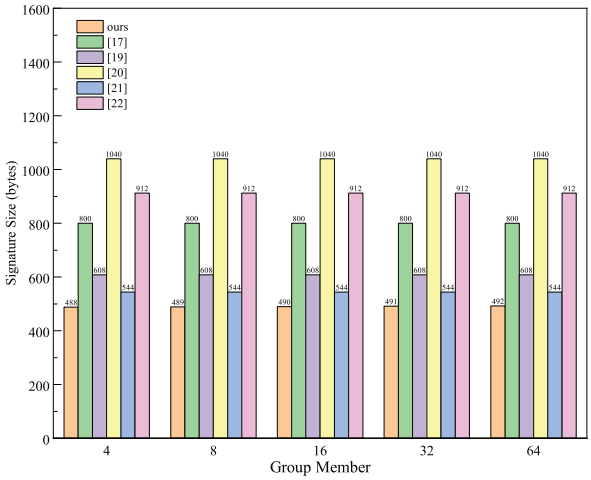


Fig. 14. Communication Cost.

D. Communication Cost

Firstly, the number of bits used in the authentication process is quantified, as shown in Table VI. Secondly, the communication cost of the PUF-CLGS is calculated and compared with the benchmark schemes [17], [19], [20], [21], and [22]. The size of our signature is given by:

$$\begin{aligned}\sigma &= |r| + |p| + |u_r| \\ &= |C| + \text{ECC point} + |\mathbb{Z}_q| + \log_2 N + a \end{aligned} \tag{31}$$

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As shown in Fig. 14, when the tree depth $a = 18$, the communication cost of the PUF-CLGS varies with the number of group members. As the number of group members increases, the communication cost of the PUF-CLGS is significantly lower than that of schemes [17], [20] and [22]. Compared with schemes [20], [22], the communication cost is similar, but the PUF-CLGS incurs lower computational cost. Overall, in resource-constrained environments, the PUF-CLGS demonstrates greater advantages in reducing both communication and computational cost, achieving higher overall efficiency and making it well-suited for post-disaster rescue scenarios.

VIII. Conclusion

To address the security and privacy issues in post-disaster rescue within drone-assisted communication scenarios, the paper proposes an application of multi-factor PUF authentication circuit based on lattice group signature in drone-assisted communication. The scheme designs a dynamic authentication strategy based on PUF-based multi-factor authentication circuit to achieve authentication and physical security under different security requirements. The scheme achieves device anonymity through lattice-based group signature and incorporates Merkle hash tree for group member updates. Security analysis shows that the proposed scheme can resist eavesdropping, impersonation, and physical clone attacks etc. Since the scheme is based on lattice-based hard problems, it can resist quantum computing attacks. Performance evaluation shows that this scheme outperforms benchmark schemes and is suitable for post-disaster rescue.

IX. Discussion and Future Work

Although the PUF-CLGS offers significant advantages over existing benchmark schemes, it still has some limitations. For example, electromagnetic interference in disaster areas may lead to a decrease in the stability of PUF circuits, energy consumption issues for drones during continuous authentication under battery-powered conditions, and a potential decline in communication performance as the number of group members increases in 5G networks or large-scale cluster networking. To address the above limitations, we will increase the number of groups to balance communication load. Meanwhile, we will investigate an electromagnetic interference-resistant PUF to enhance authentication efficiency, and explore strategies to reduce power consumption in drone operations.

Furthermore, this scheme focuses on security and privacy issues in authentication, especially the mobility of group members. It provides a detailed analysis of the potential impact on the computational or communication costs of the scheme when members join or leave the group, including analysis of the security of member list maintenance and the authentication process. However, since communication drones mainly serve as temporary communication nodes and have weak mobility, their mobility has not been explored in depth. Future work will focus on studying the efficient mobility and security guarantees of drones in dynamic environments to enhance the overall system performance.

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